



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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22MA201

ENGINEERING MATHEMATICS II

UNIT 5 –ANALYTIC FUNCTIONS

**LM1 Basic Concepts of Complex Numbers, Geometrical Representation
of Complex Number – Analytic functions and its Properties**

UNIT- V

ANALYTIC FUNCTIONS

3.1 INTRODUCTION

The theory of functions of a complex variable is the most important in solving a large number of Engineering and Science problems. Many complicated integrals of real function are solved with the help of a complex variable.

3.1 (a) Complex Variable

$x + iy$ is a complex variable and it is denoted by z .

(i. e.) $z = x + iy$ where $i = \sqrt{-1}$

3.1 (b) Function of a complex Variable

If $z = x + iy$ and $w = u + iv$ are two complex variables, and if for each value of z in a given region R of complex plane there corresponds one or more values of w is said to be a function z and is denoted by $w = f(z) = f(x + iy) = u(x, y) + iv(x, y)$ where $u(x, y)$ and $v(x, y)$ are real functions of the real variables x and y .

Note:

(i) single-valued function

If for each value of z in R there is correspondingly only one value of w , then w is called a single valued function of z .

Example: $w = z^2, w = \frac{1}{z}$

$w = z^2$					$w = \frac{1}{z}$				
z	1	2	-2	3	z	1	2	-2	3
w	1	4	4	9	w	1	$\frac{1}{2}$	$\frac{1}{-2}$	$\frac{1}{3}$

(ii) Multiple – valued function

If there is more than one value of w corresponding to a given value of z then w is called multiple – valued function.

Example: $w = z^{1/2}$

$w = z^{1/2}$			
z	4	9	1
w	-2, 2	3, -3	1, -1

(iii) The distance between two points z and z_o is $|z - z_o|$

(iv) The circle C of radius δ with centre at the point z_o can be represented by $|z - z_o| = \delta$.

(v) $|z - z_o| < \delta$ represents the interior of the circle excluding its circumference.

(vi) $|z - z_o| \leq \delta$ represents the interior of the circle including its circumference.

(vii) $|z - z_0| > \delta$ represents the exterior of the circle.

(viii) A circle of radius 1 with centre at origin can be represented by $|z| = 1$

3.1 (c) Neighbourhood of a point z_0

Neighbourhood of a point z_0 , we mean a sufficiently small circular region [excluding the points on the boundary] with centre at z_0 .

$$(i. e.) |z - z_0| < \delta$$

Here, δ is an arbitrary small positive number.

3.1 (d) Limit of a Function

Let $f(z)$ be a single valued function defined at all points in some neighbourhood of point z_0 .

Then the limit of $f(z)$ as z approaches z_0 is w_0 .

$$(i. e.) \lim_{z \rightarrow z_0} f(z) = w_0$$

3.1 (e) Continuity

If $f(z)$ is said to be continuous at $z = z_0$ then

$$\lim_{z \rightarrow z_0} f(z) = f(z_0)$$

If two functions are continuous at a point their sum, difference and product are also continuous at that point, their quotient is also continuous at any such point [$dr \neq 0$]

Example: 3.1 State the basic difference between the limit of a function of a real variable and that of a complex variable.

Solution:

In real variable, $x \rightarrow x_0$ implies that x approaches x_0 along the X-axis (or) a line parallel to the X-axis.

In complex variables, $z \rightarrow z_0$ implies that z approaches z_0 along any path joining the points z and z_0 that lie in the z -plane.

3.1 (f) Differentiability at a point

A function $f(z)$ is said to be differentiable at a point, $z = z_0$ if the limit

$$f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \text{ exists.}$$

This limit is called the derivative of $f(z)$ at the point $z = z_0$

If $f(z)$ is differentiable at z_0 , then $f(z)$ is continuous at z_0 . This is the necessary condition for differentiability.

Example: 3.2 If $f(z)$ is differentiable at z_0 , then show that it is continuous at that point.

Solution:

As $f(z)$ is differentiable at z_0 , both $f(z_0)$ and $f'(z_0)$ exist finitely.

$$\text{Now, } \lim_{z \rightarrow z_0} |f(z) - f(z_0)| = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} (z - z_0)$$

$$\begin{aligned}
&= \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \lim_{z \rightarrow z_0} (z - z_0) \\
&= f'(z_0) \cdot 0 = 0
\end{aligned}$$

Hence, $\lim_{z \rightarrow z_0} f(z) = \lim_{z \rightarrow z_0} f(z_0) = f(z_0)$

As $f(z_0)$ is a constant.

This is exactly the statement of continuity of $f(z)$ at z_0 .

3.2 ANALYTIC FUNCTIONS – NECESSARY AND SUFFICIENT CONDITIONS FOR ANALYTICITY IN CARTESIAN AND POLAR CO-ORDINATES

Analytic [or] Holomorphic [or] Regular function

A function is said to be analytic at a point if its derivative exists not only at that point but also in some neighbourhood of that point.

Entire Function: [Integral function]

A function which is analytic everywhere in the finite plane is called an entire function.

An entire function is analytic everywhere except at $z = \infty$.

Example: $e^z, \sin z, \cos z, \sinh z, \cosh z$

3.2 (i) The necessary condition for $f = (z)$ to be analytic. [Cauchy – Riemann Equations]

The necessary conditions for a complex function $f = (z) = u(x, y) + iv(x, y)$ to be analytic in a region R are $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$ i. e., $u_x = v_y$ and $v_x = -u_y$

1. Identify the function $f(z) = \bar{z}z$ is analytic at the origin.

$$f(z) = (x + iy)(x - iy) = x^2 + y^2$$

$$u + iv = x^2 + y^2$$

comparing RP and IP, $u = x^2 + y^2, v = 0$

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial x} = 0$$

$$\frac{\partial u}{\partial y} = 2y, \quad \frac{\partial v}{\partial y} = 0$$

The CREquations are satisfied at the origin..

Hence $\bar{z}z$ has a derivative only at the origin.

2. Identify the function $f(z) = z^3$ is analytic or not.

$$f(z) = (x + iy)^3 = (x^3 - 3xy^2) + i(3x^2y - y^3)$$

$$u + iv = (x^3 - 3xy^2) + i(3x^2y - y^3)$$

comparing RP and IP,

$$u = (x^3 - 3xy^2), \quad v = (3x^2y - y^3)$$

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2, \quad \frac{\partial v}{\partial x} = 6xy$$

$$\frac{\partial u}{\partial y} = -6xy, \quad \frac{\partial v}{\partial y} = 3x^2 - 3y^2$$

The CREquations are satisfied at the origin..

Hence $w = z^3$ is analytic.

3. Examine whether the following function analytic or not. $f(z) = e^x(\cos y - i \sin y)$

$$f(z) = e^x(\cos y - i \sin y)$$

$$u + iv = e^x(\cos y - i \sin y)$$

comparing RP and IP,

$$u = e^x \cos y, \quad v = -e^x \sin y$$

$$\frac{\partial u}{\partial x} = e^x \cos y, \quad \frac{\partial v}{\partial x} = -e^x \sin y$$

$$\frac{\partial u}{\partial y} = -e^x \sin y, \quad \frac{\partial v}{\partial y} = -e^x \cos y,$$

The CREquations are not satisfied.

Hence $f(z) = e^x(\cos y - i \sin y)$ is not analytic function

4. Identify the function $f(z) = \sin x \cosh y + i \cos x \sinh y$ is analytic or not

Hint: $u(x, y)$ is the velocity function (i.e. speed of the wind w.r.to time) and $v(x, y)$ is the stream line function (i.e. direction of wind flow).

$$f(z) = \sin x \cosh y + i \cos x \sinh y$$

$$u + iv = \sin x \cosh y + i \cos x \sinh y$$

comparing RP and IP,

$$u = \sin x \cosh y, \quad v = \cos x \sinh y$$

$$\frac{\partial u}{\partial x} = \cos x \cosh y, \quad \frac{\partial v}{\partial x} = -\sin x \sinh y$$

$$\frac{\partial u}{\partial y} = \sin x \sinh y, \quad \frac{\partial v}{\partial y} = \cos x \cosh y$$

The CR equations are satisfied.

Hence the function $f(z) = \sin x \cosh y + i \cos x \sinh y$ is analytic

5. Identify the function $f(z) = (x^2 - y^2 + 2xy) + i(x^2 - y^2 - 2xy)$ is analytic or not

$$f(z) = (x^2 - y^2 + 2xy) + i(x^2 - y^2 - 2xy)$$

$$u + iv = (x^2 - y^2 + 2xy) + i(x^2 - y^2 - 2xy)$$

comparing RP and IP,

$$u = (x^2 - y^2 + 2xy), \quad v = (x^2 - y^2 - 2xy)$$

$$\frac{\partial u}{\partial x} = 2x + 2y, \quad \frac{\partial v}{\partial x} = 2x - 2y$$

$$\frac{\partial u}{\partial y} = -2y + 2x, \quad \frac{\partial v}{\partial y} = -2y - 2x$$

The CR equations are not satisfied

Hence the function $f(z) = (x^2 - y^2 + 2xy) + i(x^2 - y^2 - 2xy)$ is not analytic

6. Identify the function $f(z) = \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1} \left[\frac{y}{x} \right]$ is analytic or not

$$f(z) = \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1} \left[\frac{y}{x} \right]$$

$$u + iv = \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1} \left[\frac{y}{x} \right]$$

comparing RP and IP,

$$u = \frac{1}{2} \log(x^2 + y^2), \quad v = \tan^{-1} \left[\frac{y}{x} \right]$$

$$\frac{\partial u}{\partial x} = \frac{x}{x^2 + y^2}, \quad \frac{\partial v}{\partial x} = \frac{-y}{x^2 + y^2}$$

$$\frac{\partial u}{\partial y} = \frac{y}{x^2 + y^2}, \quad \frac{\partial v}{\partial y} = \frac{x}{x^2 + y^2}$$

The CREquations are satisfied

Hence $f(z) = \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1} \left[\frac{y}{x} \right]$ is analytic

7. Show that the function $u(x, y) = 3x^2y + 2x^2 - y^3 - 2y^2$ is harmonic function

$$u(x, y) = 3x^2y + 2x^2 - y^3 - 2y^2$$

$$\frac{\partial u}{\partial x} = 6xy + 4x$$

$$\frac{\partial u}{\partial y} = 3x^2 - 3y^2 + 4y$$

$$\frac{\partial^2 u}{\partial x^2} = 6y + 4$$

$$\frac{\partial^2 u}{\partial y^2} = -6y + 4$$

$$\text{Therefore } \nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6y + 4 - 6y - 4 = 0.$$

Hence the given function $u(x, y) = 3x^2y + 2x^2 - y^3 - 2y^2$ is harmonic.

8. Show that the function $u(x, y) = xy(x^2 - y^2)$ is harmonic function

$$u(x, y) = xy(x^2 - y^2) = x^3y - xy^3$$

$$\frac{\partial u}{\partial x} = 3x^2y - y^3$$

$$\frac{\partial u}{\partial y} = x^3 - 3xy^2$$

$$\frac{\partial^2 u}{\partial x^2} = 6xy$$

$$\frac{\partial^2 u}{\partial y^2} = -6xy$$

$$\text{Therefore } \nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6xy - 6xy = 0.$$

9. Show that the function $u(x, y) = (x - y)(x^2 + 4xy + y^2)$ is harmonic function

$$u(x, y) = (x - y)(x^2 + 4xy + y^2)$$

$$\frac{\partial u}{\partial x} = (x - y)(2x + 4y) + (x^2 + 4xy + y^2)$$

$$\frac{\partial u}{\partial y} = (x - y)(4x + 2y) + (x^2 + 4xy + y^2) \quad (-1)$$

$$\frac{\partial^2 u}{\partial x^2} = (x - y)(2) + (2x + 4y)(1) + (2x + 4y) = 6x + 6y$$

$$\frac{\partial^2 u}{\partial y^2} = (-1)(4x + 2y) + (x - y)(2) - 4x - 2y = -6x - 6y$$

$$\text{Therefore } \nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x + 6y - 6x - 6y = 0.$$

DISCUSSION QUESTIONS:

1. What is the concept of a complex potential in fluid flow?
2. How are analytic functions utilized in various branches of science and engineering?
3. Application of analytic function in fluid flow?